

4.5.2 Chemical quantities

This topic shows how chemists use quantitative methods to determine the formulae of compounds and the equations for reactions. Given this information, they can determine reacting quantities, assess the purity of products and monitor the yield from chemical reactions. The methods can be developed and applied in the context of other chemical topics in this specification. Note that arithmetic computation, ratio, percentage and multi-step calculations feature throughout this section.

4.5.2.1 Chemical equations

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Use the names and symbols of common elements and compounds and the principle of conservation of mass to write formulae and balanced chemical equations. (HT only) and half equations. Use chemical symbols to write the formulae of elements and simple covalent and ionic compounds. Describe the physical states of products and reactants using state symbols (s, l, g and aq).	Atoms of each element are represented by a chemical symbol, eg O represents an atom of oxygen, Na represents an atom of sodium. There are about 100 different elements. Elements are shown in the periodic table. Compounds are formed from elements by chemical reactions. Compounds contain two or more elements chemically combined in fixed proportions and can be represented by formulae using the symbols of the atoms from which they were formed. Compounds can only be separated into elements by chemical reactions. Chemical reactions always involve the formation of one or more new substances, and often involve a detectable energy change. Chemical reactions can be represented by word equations or equations using symbols and formulae. In chemical equations, the three states of matter are shown as (s), (l) and (g), with (aq) for aqueous solutions.	WS 4.1 Use the names and symbols of the first 20 elements, groups 1, 7 and 0 and other common elements from a supplied periodic table to write formulae and balanced chemical equations where appropriate. Name compounds of these elements from given formulae or symbol equations. Write word equations for the reactions in this specification. Write formulae and balanced chemical equations for the reactions in this specification.

4.5.2.2 Conservation of mass

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall and use the law of conservation of mass.	The law of conservation of mass states that no atoms are lost or made during a chemical reaction so the mass of the products equals the mass of the reactants. This means that chemical reactions can be represented by symbol equations that are balanced in terms of the numbers of atoms of each element involved on both sides of the equation.	MS 1a Use arithmetic computation and ratio when writing and balancing equations.

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain any observed changes in mass in non-enclosed systems during a chemical reaction and explain them using the particle model.	Some reactions may appear to involve a change in mass but this can usually be explained because a reactant or product is a gas and its mass has not been taken into account.	WS 1.2 Explain any observed changes in mass in non-enclosed systems during a chemical reaction given the balanced symbol equation for the reaction.

4.5.2.3 Relative formula masses

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Calculate relative formula masses of species separately and in a balanced chemical equation. Students should be able to calculate the percentage by mass in a compound given the relative formula mass and the relative atomic masses.	The relative atomic mass of an element compares the mass of atoms of the element with the ^{12}C isotope. It is an average value for the isotopes of the element. The relative formula mass (M_r) of a compound is the sum of the relative atomic masses of the atoms in the numbers shown in the formula. In a balanced chemical equation, the sum of the relative formula masses of the reactants in the quantities shown equals the sum of the relative formula masses of the products in the quantities shown. Students will not be expected to calculate relative atomic masses from isotopic abundances.	MS 1a, 3a Calculate the relative formula mass (M_r) of a compound from its formula, given the relative atomic masses. WS 3.3 Carry out and represent mathematical and statistical analysis.

4.5.2.4 Amounts in moles (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the mass of a given substance is related to the amount of that substance in moles and vice versa. Recall and use the definitions of the Avogadro constant (in standard form) and of the mole.	Chemical amounts are measured in moles. The symbol for the unit mole is mol. The mass of one mole of a substance in grams is numerically equal to its relative formula mass. One mole of a substance contains the same number of the stated particles, atoms, molecules or ions as one mole of any other substance. The number of atoms, molecules or ions in a mole of a given substance is the Avogadro constant. The value of the Avogadro constant is 6.02×10^{23} per mole.	MS 1a, 1b, 1c, 2a Recognise and use expressions in decimal form when using the relative formula mass of a substance to calculate the amount in moles in a given mass of that substance and vice versa, giving the answer in the appropriate units. MS 3b, 3c Change the subject of a mathematical equation. WS 4.6, MS 2a Provide answers to an appropriate number of significant figures. MS 1b Calculate with numbers written in standard form when using the Avogadro constant. MS 3a Understand and use the symbols: =, <, <<, >>, >, $\propto$, ~

4.5.2.5 Calculations based on equations (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Deduce the stoichiometry of an equation from the masses of reactants and products and explain the effect of a limiting quantity of a reactant.	The balancing numbers in a symbol equation can be calculated from the masses of reactants and products by converting the masses in grams to amounts in moles and converting the numbers of moles to simple whole number ratios. In a chemical reaction involving two reactants, it is common to use an excess of one of the reactants to ensure that all of the other reactant is used. The reactant that is completely used up is called the limiting reactant.	MS 3c, 3d Balance an equation given the masses of reactants and products. Explain the effect of a limiting quantity of a reactant on the amount of products it is possible to obtain in terms of amounts in moles or masses in grams.
Use a balanced equation to calculate masses of reactants or products.	The masses of reactants and products can be calculated from balanced symbol equations. Chemical equations can be interpreted in terms of moles. For example: $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$ shows that one mole of magnesium reacts with two moles of hydrochloric acid to produce one mole of magnesium chloride and one mole of hydrogen gas.	MS 1a, 1c, 3c, 3d Calculate the masses of reactants and products from the balanced symbol equation and the mass of a given reactant or product. WS 4.6, MS 2a Provide answers to an appropriate number of significant figures.

4.5.2.6 Concentrations of solutions

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
(HT only) Explain how the mass of a solute and the volume of the solution is related to the concentration of the solution.	Many chemical reactions take place in solutions. The concentration of a solution can be measured in mass per given volume of solution, eg grams per dm³ (g/dm³).	MS 1c, 3c Calculate the mass of solute in a given volume of solution of known concentration in terms of mass per given volume of solution.